

Spontaneous Parametric Down Conversion in nonlinear photonic crystals

SH1012 – Project Report

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Background and Theory

Photonics

Atomic Excitation & Photon Pair Production (Harris, kursbok)

Nämna Maxwell's eq --> Wave Equation? (Boyd)

Destructive Wave Interference (Tänker att detta förklarar varför det är viktigt med periodic poling)

Spontaneous Parametric Down Conversion (SPDC) (Bra avsnitt om detta i Boyd kap 2 – Difference-Frequency Generation)

En bra källa att peka på här är nog Harris et al. 1967 (Första observationen av SPDC)

Material Properties (Boyd & artiklar?)

Electromagnetic Polarization

Maxwell's eq --> Wave Equation? (Boyd)

Boyd kapitel 2 behandlar i ganska bra detalj hur Maxwells lagar fungerar i icke-linjära optiska medium, vilket leder in i polarisation av materialet och hur vi kan betrakta $P(r, t)$ som en summa av frekvenskomponenter. Vilket leder till χ^2

Higher Order Susceptability (χ^2)

Refractive Index

The Sellmeier Equation (Vi behöver nog källa på denna, alternativt att vi visar matematiskt hur ekvationen för vissa våglängder bli komplex och/eller mindre än 1)

Sellmeier Equation breakdown at wavelengths approaching infrared (spottar ut ofysikaliskt n ($n < 1$) vid vissa våglängder och då $n \in \mathbb{C}$ absorberar materialet EM-strålning.

Periodic Poling

Quasi Phase Matching & The Reciprocal Lattice **G**

Method

Conservation of energy

$\lambda_i(\lambda_s, \lambda_p)$ --> Signal Wavelength --> Wavelength singularity

Momentum Conservation (1D)

Quasi Phase Matching

System of Equations (Visa matten)

Conservation of energy:

$$E = h \cdot f$$

$$f = \frac{\omega}{2\pi}$$

$$\therefore E_p = E_s + E_i = \omega_p = \omega_s + \omega_i \quad (1)$$

$$\omega = 2\pi \cdot f = \frac{2\pi}{\lambda} \cdot c_0 \quad (2)$$

$$(1), (2) \Rightarrow \frac{1}{\lambda_p} = \frac{1}{\lambda_s} + \frac{1}{\lambda_i} \quad (3)$$

Fixing signal wavelength directly determines idler wavelength:

$$(3) \Rightarrow \lambda_i(\lambda_p, \lambda_s) = \frac{\lambda_s \lambda_p}{\lambda_s - \lambda_p} \quad (4)$$

Note that (4) diverges when $\lambda_s \rightarrow \lambda_p$, singularity in numerical model.

The Wave Vector norm $|k|$:

For a given angular frequency ω_m , the wave vector norm is defined as a function of the refractive index $n(T, \omega_m)$, given by the Sellmeier equation, the speed of the wave (c_0) and the angular frequency which we rewrite using (2):

$$k_m = |k(T, \omega_m)| = \frac{\omega_m}{c_0} n(T, \omega_m) = \frac{2\pi}{\lambda_m} n(T, \lambda_m) \quad (5)$$

As mentioned in the section on destructive interference, to achieve phase matching we are looking for a system where:

$$\Delta k = k_p - k_s - k_i = 0$$

As noted in the section on material however, this is rarely achievable due to deconstructive wave interference, which is why we introduce a correction in the form of the reciprocal lattice vector G :

$$\Delta k = k_p - k_s - k_i - G \approx 0, \text{ where } G = \frac{1}{\Lambda}$$

Now, considering the 2-dimensional model (figure 1), we project the problem onto the plane which reduces the conservation of momentum and therefore the phase matching equations (excluding the reciprocal lattice vector for now) to two simple trigonometric expressions:

$$\text{Along the optical axis: } k_p = k_s \cdot \cos(\theta_s) + k_i \cdot \cos(\theta_i) \quad (6)$$

$$\text{Perpendicular to the optical axis: } 0 = k_s \cdot \sin(\theta_s) - k_i \cdot \sin(\theta_i) \quad (7)$$

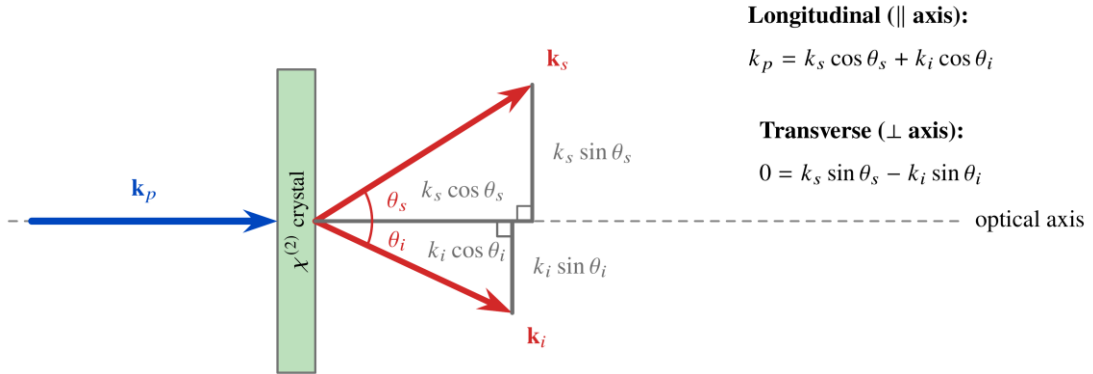


Figure 1: 2-dimensional model of the SPDC geometry.

Rearranging (7) we arrive at an expression for the idler angle as a function of the signal angle:

$$\theta_i = \arcsin\left(\frac{k_s}{k_i} \cdot \sin(\theta_s)\right) \quad (8)$$

Finally, combining (4), (5), (6), (8) and adding the reciprocal lattice vector to (6) we arrive at a root finding function which can be used to find angles and wavelengths that fulfill both conservation of- energy and -momentum conditions that we expect a SPDC to fulfill:

$$f(\theta_s, \lambda_s) = k_p - k_s \cdot \cos(\theta_s) - k_i \cos(\theta_i) - G = 0 \quad (9)$$

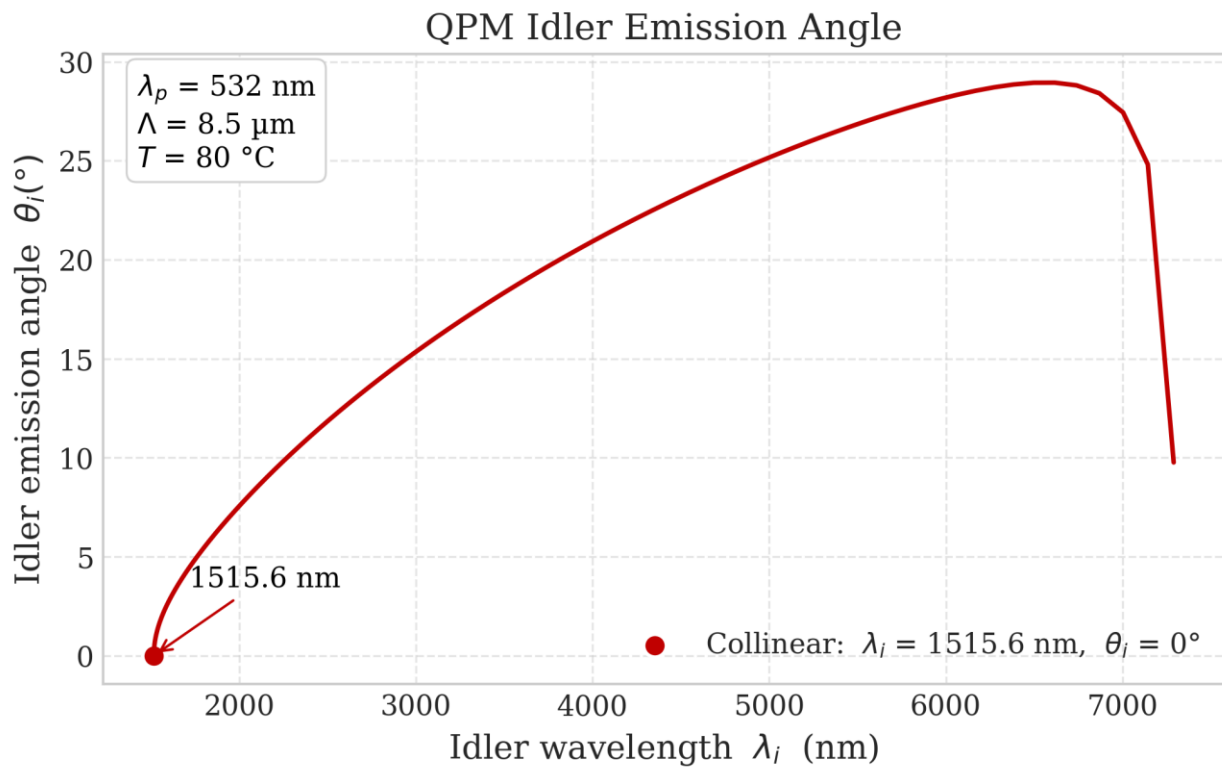
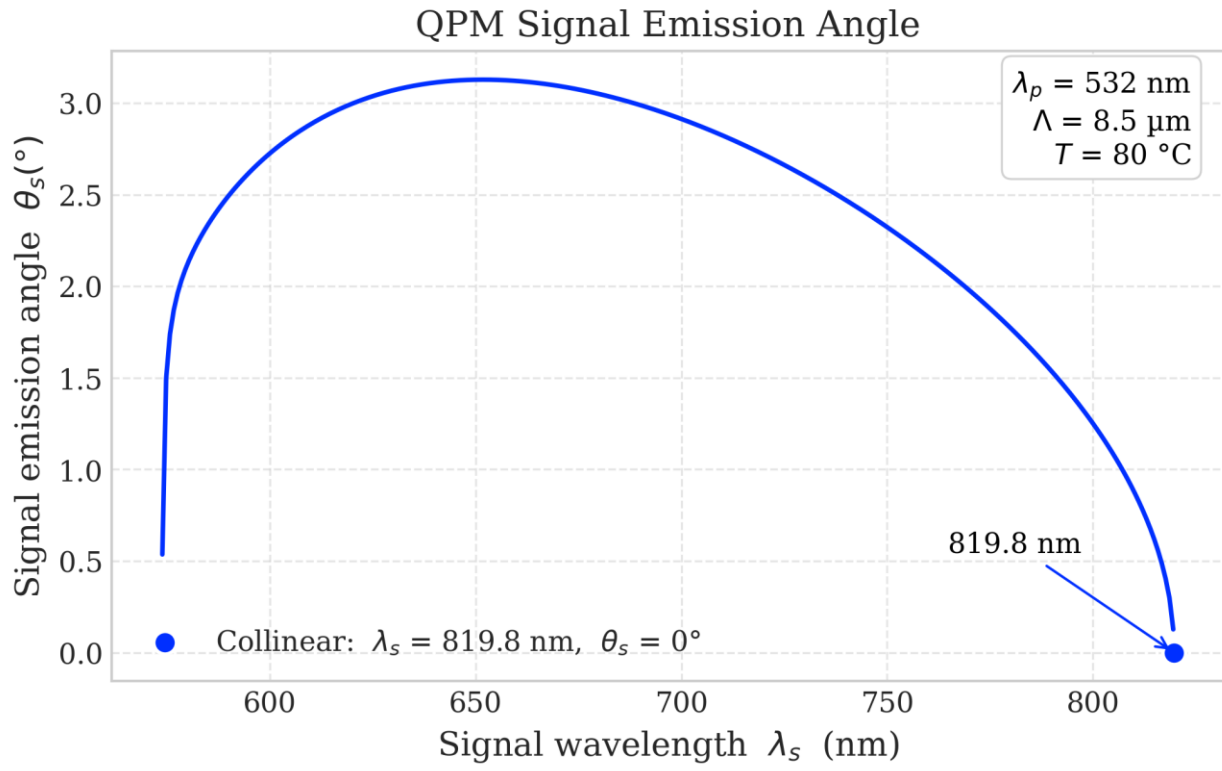
Brent's Method, a root finding method (numeriska metoden vi använt)

Results

At $\theta_s = 1.0^\circ$: $\lambda_s = 807.49$ nm, $\lambda_i = 1559.33$ nm, $\theta_i = 1.935^\circ$

λ_s (nm)	λ_i (nm)	n_s	n_i	θ_s (°)	θ_i (°)
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Collinear phase-matching wavelength: signal: 819.75 nm, idler: 1515.57 nm



Discussion